



Edge AI: Test Specifications Document

Version 1

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OTAInfo ML Model Workflow Test – Simple Setup

Objective

Evaluate the performance of the OTAInfo outcome predictor model in a non-specialized, personal environment. This version of the test is designed to work without lab-grade tools, enabling individual users to replicate efficiency validation of the model.

Background

The OTAInfo model has been tested in lab conditions, and its performance is documented in the model card and related publications. This iteration allows individual users to verify model behavior in real-world Wi-Fi environments using a basic ESP32 setup.

Pre-requisites

- **ESP32 board** connected to your laptop via USB
 - **Wi-Fi router** (standard home/office router)
 - [Optional] If using prebuilt **.bin** firmware images:
 - Wi-Fi SSID must be **test_iot**
 - Password must be **Test123\$56**
 - [Optional] Local setup with **ESP-IDF** and **TensorFlow Lite for Microcontrollers** to build from source
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Step 1: Wi-Fi Signal Mapping

1. Flash your ESP32 with the **wifi_rssi_mapper** firmware.
2. Open a serial monitor (e.g., **idf.py monitor** or use serial tool).
3. Walk away from the router slowly, observing RSSI values on the monitor.

4. Identify a location where the RSSI is stable between **-70 dBm and -80 dBm**.
 5. Stand at that spot for **2 minutes** to confirm signal stability.
 6. **Mark this position** (e.g., with tape or an object); this will be your **test location** for the next steps.
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Step 2: Regular Ping Baseline

1. Flash the ESP32 with the **wifi_ping_test** firmware.
 2. Go to the previously marked **test location**.
 3. Start monitoring via serial connection.
 4. Run the ping test and let it complete one session.
 5. Repeat the test **three times**, disconnecting the ESP32 for 1 minute between each run.
 6. After each run, copy the results to a Notepad, Google Doc, or spreadsheet for comparison.
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Step 3: AI-Powered Ping Test

1. **Flash the ESP32** with the **otainfo_ping_test** firmware.
2. **Return to the marked test location** identified during the Wi-Fi signal mapping step.
3. **Connect the ESP32 to your laptop** and open a serial monitor (e.g., **idf.py monitor**).
4. **Initiate the ping test** and allow it to complete one full session.
5. **Repeat the test three times**, waiting 1 minute between each run. Disconnect and reconnect the ESP32 before each new run to simulate fresh startup conditions.

6. **Record the output** after each session by copying the results into a Notepad, Google Doc, or spreadsheet for later comparison with the regular ping test.
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 Ping Test Results Comparison

Run #	Test Type	Packets Sent	Packets Received	Packet Loss (%)	Average Latency (ms)	Comments
1	Regular Ping	5	1	80%	68	Only first ping responded
2	Regular Ping	5	1	80%	45	Similar poor response
3	Regular Ping	5	0	100%	-	WiFi reconnection issues, panic error
1	AI Ping (OTAInfo)	5	5	0%	52.4	Stable across all packets
2	AI Ping (OTAInfo)	5	4	20%	56.25	One packet lost
3	AI Ping (OTAInfo)	5	5	0%	99.8	One high-latency packet (285ms)

 Notes:

- **Average Latency Calculation (AI):** Rounded mean of the individual ping latencies.
- **Test Location:** As per instructions, likely around -70 to -80 dBm signal.
- **Observation:** The AI-powered ping tests clearly outperformed regular pings in both reliability and success rate, even under challenging signal conditions.
- **Regular Ping Run #3** failed due to an exception (**IntegerDivideByZero**), making it unusable for practical purposes.

Appendix

RAW data Regular ping:

[Test #1]

```
I (7934) wifi:dp: 1, bi: 102400, li: 3, scale listen interval from 307200 us to 307200 us
I (7944) wifi:AP's beacon interval = 102400 us, DTIM period = 3
I (9444) wifi:<ba-add>idx:0 (ifx:0, 56:ab:f8:52:6d:12), tid:0, ssn:0, winSize:64
I (10464) esp_netif_handlers: sta ip: 192.168.0.157, mask: 255.255.255.0, gw: 192.168.0.1
I (10464) ping_example: Connected to SSID: test_iot
I (12464) main_task: Returned from app_main()
I (12534) ping_example: 64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=68 ms
W (14464) ping_example: Request timeout for icmp_seq 2
W (15464) ping_example: Request timeout for icmp_seq 3
W (16464) ping_example: Request timeout for icmp_seq 4
W (17464) ping_example: Request timeout for icmp_seq 5
I (17464) ping_example: --- Ping statistics ---
I (17464) ping_example: 5 packets transmitted, 1 received, 80% packet loss
```

[Test #2]

```
I (13294) wifi:dp: 1, bi: 102400, li: 3, scale listen interval from 307200 us to 307200 us
I (13304) wifi:AP's beacon interval = 102400 us, DTIM period = 3
I (13314) wifi:<ba-add>idx:0 (ifx:0, 56:ab:f8:52:6d:12), tid:0, ssn:0, winSize:64
I (14344) esp_netif_handlers: sta ip: 192.168.0.157, mask: 255.255.255.0, gw: 192.168.0.1
I (14344) ping_example: Connected to SSID: test_iot
I (16344) main_task: Returned from app_main()
I (16394) ping_example: 64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=45 ms
W (18344) ping_example: Request timeout for icmp_seq 2
W (19344) ping_example: Request timeout for icmp_seq 3
W (20344) ping_example: Request timeout for icmp_seq 4
W (21344) ping_example: Request timeout for icmp_seq 5
I (21344) ping_example: --- Ping statistics ---
I (21344) ping_example: 5 packets transmitted, 1 received, 80% packet loss
```

[Test #3]

```
I (884) wifi:dp: 1, bi: 102400, li: 3, scale listen interval from 307200 us to 307200 us
I (894) wifi:AP's beacon interval = 102400 us, DTIM period = 3
I (984) wifi:<ba-add>idx:0 (ifx:0, 56:ab:f8:52:6d:12), tid:0, ssn:0, winSize:64
I (2044) esp_netif_handlers: sta ip: 192.168.0.157, mask: 255.255.255.0, gw: 192.168.0.1
I (2044) ping_example: Connected to SSID: test_iot
I (4014) wifi:state: run -> init (0x22c0)
I (4034) wifi:pm stop, total sleep time: 1200916 us / 3147297 us
I (4034) wifi:<ba-del>idx:0, tid:0
I (4034) wifi:new:<6,0>, old:<6,1>, ap:<255,255>, sta:<6,1>, prof:1, snd_ch_cfg:0x0
I (4044) ping_example: Retrying Wi-Fi...
E (4044) ping_sock: send error=0
I (4044) main_task: Returned from app_main()
I (4074) wifi:new:<6,1>, old:<6,0>, ap:<255,255>, sta:<6,1>, prof:1, snd_ch_cfg:0x0
I (4074) wifi:state: init -> auth (0xb0)
I (4104) wifi:state: auth -> assoc (0x0)
W (5044) ping_example: Request timeout for icmp_seq 1
```

```

E (5044) ping_sock: send error=0
I (5104) wifi:state: assoc -> init (0x400)
I (5114) wifi:new:<6,0>, old:<6,1>, ap:<255,255>, sta:<6,1>, prof:1, snd_ch_cfg:0x0
I (5114) ping_example: Retrying Wi-Fi...
W (6044) ping_example: Request timeout for icmp_seq 2
E (6044) ping_sock: send error=0
W (7044) ping_example: Request timeout for icmp_seq 3
E (7044) ping_sock: send error=0
I (7534) ping_example: Retrying Wi-Fi...
I (7554) wifi:new:<6,1>, old:<6,0>, ap:<255,255>, sta:<6,1>, prof:1, snd_ch_cfg:0x0
I (7554) wifi:state: init -> auth (0xb0)
I (7584) wifi:state: auth -> assoc (0x0)
I (7594) wifi:state: assoc -> run (0x10)
I (7624) wifi:connected with test_iot, aid = 1, channel 6, 40U, bssid = 56:ab:f8:52:6d:12
I (7624) wifi:security: WPA2-PSK, phy: bgn, rssi: -69
I (7624) wifi:pm start, type: 1
I (7644) wifi:<ba-add>idx:0 (ifx:0, 56:ab:f8:52:6d:12), tid:0, ssn:0, winSize:64
I (7654) wifi:AP's beacon interval = 102400 us, DTIM period = 3
W (8044) ping_example: Request timeout for icmp_seq 4
E (8044) ping_sock: send error=0
I (8684) esp_netif_handlers: sta ip: 192.168.0.157, mask: 255.255.255.0, gw: 192.168.0.1
W (9044) ping_example: Request timeout for icmp_seq 5
I (9044) ping_example: --- Ping statistics ---
Guru Meditation Error: Core 0 panic'ed (IntegerDivideByZero). Exception was unhandled.
Core 0 register dump:
PC      : 0x400d8b39 PS      : 0x00060530 A0      : 0x800e2d23 A1      : 0x3ffc7dd0
--- 0x400d8b39: on_ping_end at /Users/Akshay/esp/esp-idf/examples/wifi_ping_test/main/wifi_ping_test.c:114 (discriminator
1)
A2      : 0x3ffc6d4c A3      : 0x00000000 A4      : 0x3ffb0120 A5      : 0x3ffc6ef4
A6      : 0x00000003 A7      : 0x3f403aa8 A8      : 0x00000000 A9      : 0x00000000
A10     : 0x00002354 A11     : 0x3f403aa8 A12     : 0x3f403b08 A13     : 0x00002354
A14     : 0x3f403aa8 A15     : 0x00000000 SAR     : 0x00000004 EXCCAUSE: 0x00000006
EXCVADDR: 0x00000000 LBEG    : 0x400014fd LEND    : 0x4000150d LCOUNT    : 0xffffffff
--- 0x400014fd: strlen in ROM
--- 0x4000150d: strlen in ROM
Backtrace: 0x400d8b36:0x3ffc7dd0 0x400e2d20:0x3ffc7e10 0x40089531:0x3ffc7e60
--- 0x400d8b36: on_ping_end at /Users/Akshay/esp/esp-idf/examples/wifi_ping_test/main/wifi_ping_test.c:114 (discriminator
1)
--- 0x400e2d20: esp_ping_thread at /Users/Akshay/esp/esp-idf/components/lwip/apps/ping/ping_sock.c:192
--- 0x40089531: vPortTaskWrapper at
/Users/Akshay/esp/esp-idf/components/freertos/FreeRTOS-Kernel/portable/xtensa/port.c:139
ELF file SHA256: dadb99665

```

RAW data AI ping:

[AI TEST#1]

Checking resources status..

Score: -5

Score: -5

I (3831) ping_example: 64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=40 ms

Score: -5

Score: -5

I (4871) ping_example: 64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=88 ms

Score: -5

Score: -5

I (5831) ping_example: 64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=46 ms

Score: -5

Score: -5

I (6811) ping_example: 64 bytes from 8.8.8.8: icmp_seq=4 ttl=112 time=27 ms

Score: -5

Score: -5

I (7851) ping_example: 64 bytes from 8.8.8.8: icmp_seq=5 ttl=112 time=61 ms

Score: -5

I (8791) ping_example: --- Ping statistics ---

I (8791) ping_example: 5 packets transmitted, 5 received, 0% packet loss

Score: -5

I (8831) wifi:sta rx csa, newchan=1, old=6

I (9141) wifi:sta rx csa, newchan=1, old=6

Score: -5

I (9431) wifi:switch to channel 1

I (9431) wifi:new:<1,0>, old:<6,1>, ap:<255,255>, sta:<1,0>, prof:1, snd_ch_cfg:0x0

I (9551) wifi:new:<1,1>, old:<1,0>, ap:<255,255>, sta:<1,1>, prof:1, snd_ch_cfg:0x0

Score: -5

Score: -5

[AI TEST#2]

I (5801) ping_example: Connected to SSID: test_iot

Checking resources status..

Score: -5

Score: -5

I (7861) ping_example: 64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=61 ms

Score: -5

Score: -5

I (8861) ping_example: 64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=59 ms

Score: -5

Score: -5

I (9851) ping_example: 64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=57 ms

Score: -5

Score: -5

I (10841) ping_example: 64 bytes from 8.8.8.8: icmp_seq=4 ttl=112 time=48 ms

Score: -5

Score: -5

Score: -5

W (12801) ping_example: Request timeout for icmp_seq 5

I (12801) ping_example: --- Ping statistics ---

I (12801) ping_example: 5 packets transmitted, 4 received, 20% packet loss

Score: -5

Score: -5

[AI TEST#3]

I (15851) wifi:dp: 1, bi: 102400, li: 3, scale listen interval from 307200 us to 307200 us
I (15871) wifi:AP's beacon interval = 102400 us, DTIM period = 3
I (15891) wifi:<ba-add>idx:0 (ifx:0, 56:ab:f8:52:6d:12), tid:0, ssn:0, winSize:64
I (17111) esp_netif_handlers: sta ip: 192.168.0.157, mask: 255.255.255.0, gw: 192.168.0.1
I (17111) ping_example: Connected to SSID: test_iot
Checking resources status..
Score: -5
Score: -5
I (19181) ping_example: 64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=71 ms
Score: -5
Score: -5
I (20391) ping_example: 64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=285 ms
Score: -5
Score: -5
I (21151) ping_example: 64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=48 ms
Score: -5
Score: -5
I (22161) ping_example: 64 bytes from 8.8.8.8: icmp_seq=4 ttl=112 time=52 ms
Score: -5
Score: -5
I (23151) ping_example: 64 bytes from 8.8.8.8: icmp_seq=5 ttl=112 time=43 ms
Score: -5
I (24111) ping_example: --- Ping statistics ---
I (24111) ping_example: 5 packets transmitted, 5 received, 0% packet loss